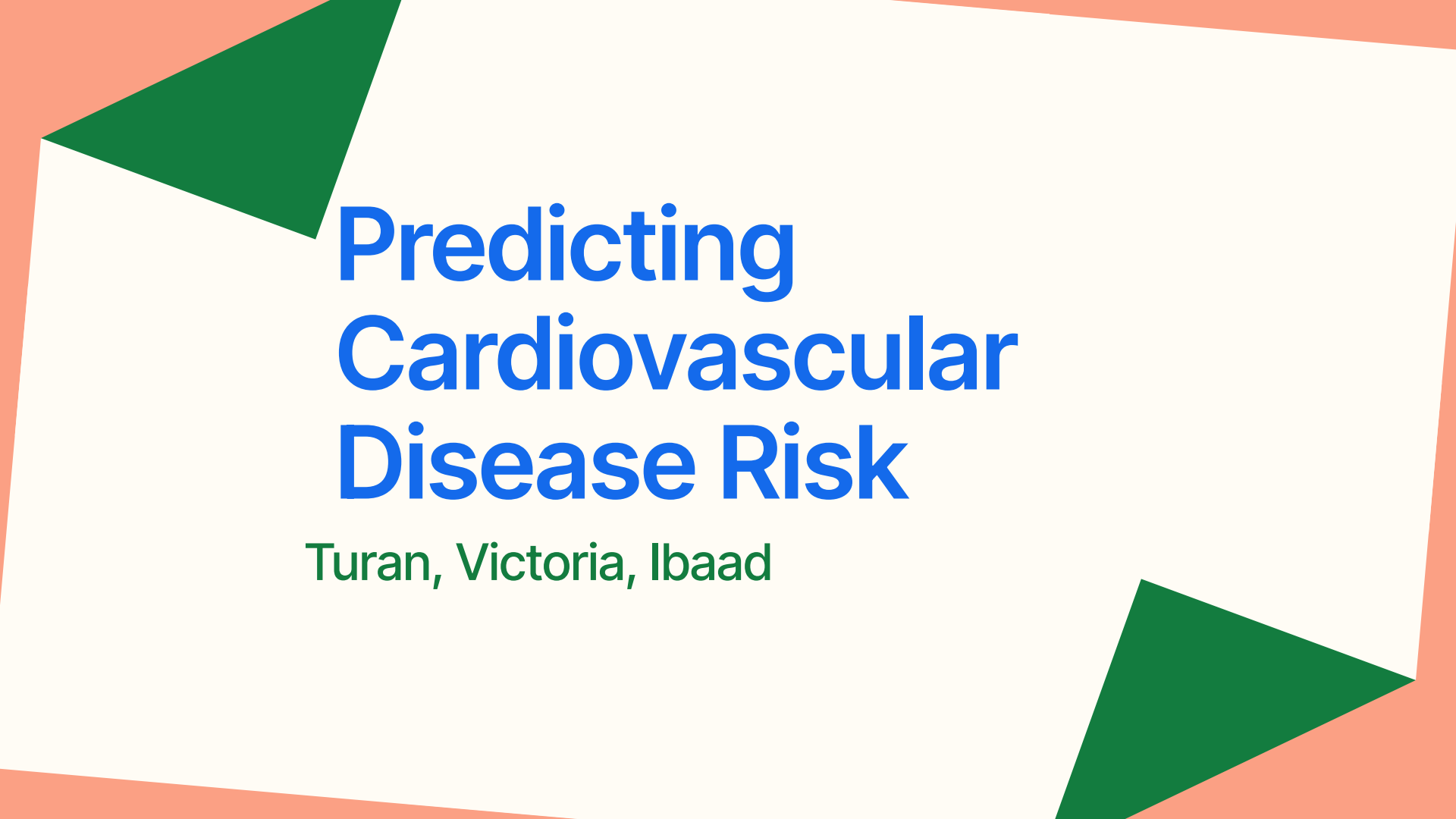


The slide features a light orange background with two large, dark green triangles. One triangle is in the top-left corner, and the other is in the bottom-right corner. The main title is centered in a bold blue font.

Predicting Cardiovascular Disease Risk

Turan, Victoria, Ibaad

Project Objective

The goal of this project is to develop a machine learning classification model that predicts the severity of cardiovascular disease risk using patient diagnostic data.

Dataset Introduction

Dataset contains cardiovascular disease risk assessment data collected in Bangladesh from January 20, 2024 to January 1, 2025

Key characteristics:

- 1,529 patient records
- 1,529 patient records
- The dataset includes demographic information, anthropometric measurements, clinical readings, biochemical markers, and lifestyle factors

Target Variable:

- CVD Risk Level
 - High
 - Intermediary
 - Low

Why This Dataset



Cardiovascular disease (CVD) is the leading cause of death among women worldwide.

However,

- Symptoms of heart disease often present differently in women
- Women have historically been underrepresented in cardiovascular research
- Leads to misdiagnosis and delayed treatment

Understanding patterns in cardiovascular risk factors can help improve early detection and prevention strategies.

Exploratory Data Analysis (EDA)

EDA was conducted to understand:

- Data structure
- Variable distributions
- Missing values / Outliers
- Relationships between predictors
- Class distribution of the target variable

	missing_count	missing_pct
Diastolic BP	82	5.36
Weight (kg)	81	5.30
HDL (mg/dL)	80	5.23
Waist-to-Height Ratio	79	5.17
Age	78	5.10
Height (cm)	74	4.84
Total Cholesterol (mg/dL)	73	4.77
Systolic BP	71	4.64
CVD Risk Score	70	4.58
Estimated LDL (mg/dL)	69	4.51
Height (m)	67	4.38
Abdominal Circumference (cm)	67	4.38
Fasting Blood Sugar (mg/dL)	67	4.38
BMI	64	4.19

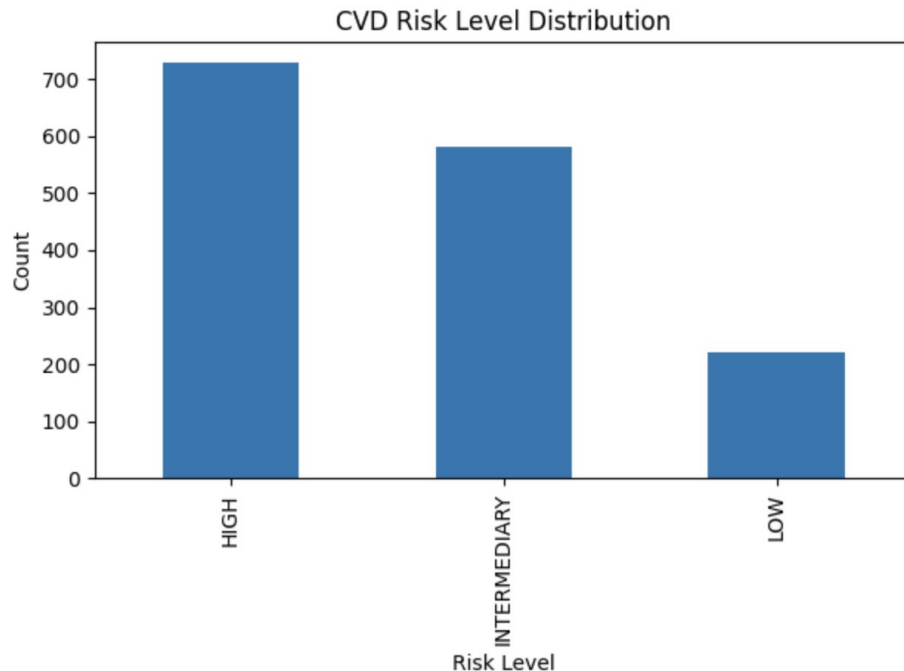
Target Variable Distribution

Distribution of CVD Risk Levels

The target variable represents the severity of cardiovascular disease risk.

CVD risk categories are distributed as follows:

- High Risk: 728 patients
- Intermediary Risk: 581 patients
- Low Risk: 220 patients



Distribution of Numeric Features

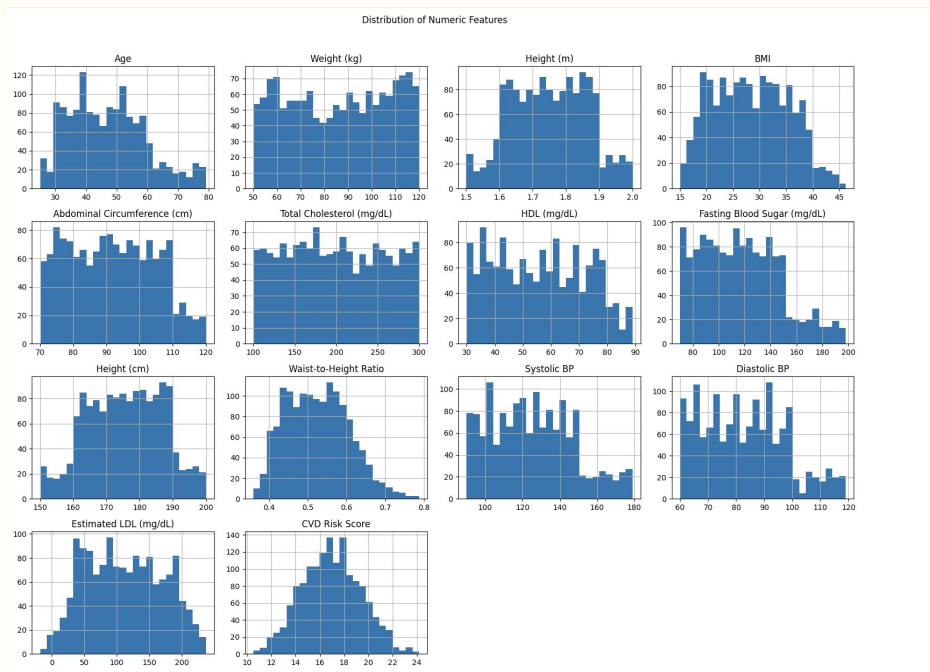
Histograms were used to show the distributions of key health indicators in the dataset that are numeric

Helps identify:

- Outliers
- Skewed distributions
- Potential feature scaling needs before modeling.

Key Observations:

- Some indicators (such as cholesterol and LDL) display longer tails, suggesting possible outliers.
- Most numeric features don't follow a specific "skew" except for the CVD risk score which appears Gaussian



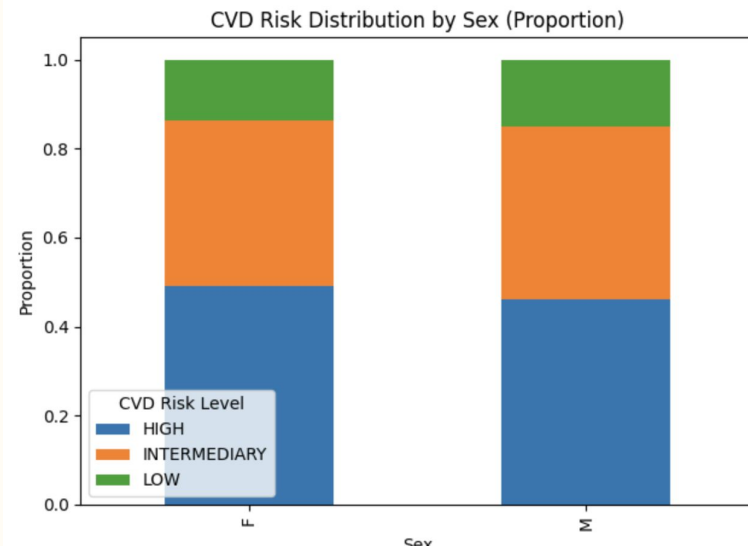
Gender Based Analysis

The distribution of CVD risk was examined by sex.

Risk distribution:

- Female Patients
 - High risk: 49.0%
 - Intermediary: 37.3%
 - Low risk: 13.7%
- Male Patients
 - High risk: 46.2%
 - Intermediary: 38.8%
 - Low risk: 15.1%

Risk distributions are similar across sexes, though female patients show a slightly higher proportion of high-risk cases.



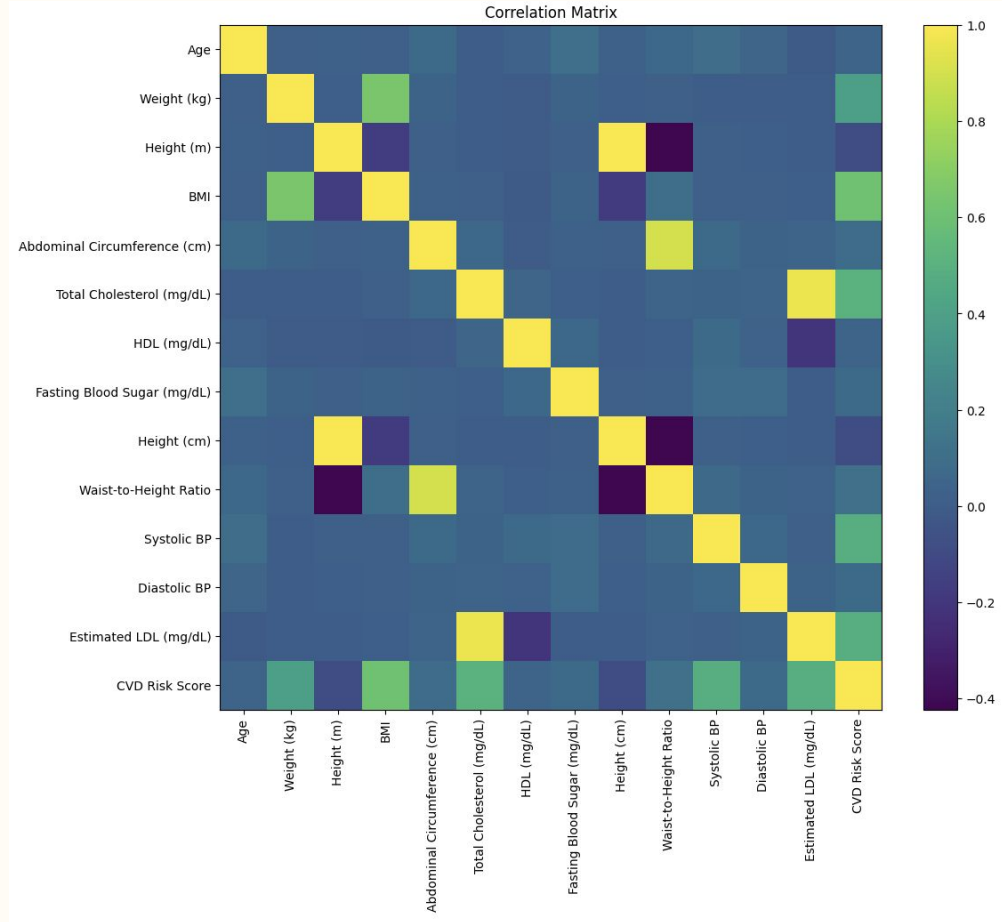
Correlation Analysis

The correlation matrix helps identify:

- Relationships between cardiovascular health indicators
- Variables that may provide similar information
- Potential multicollinearity among predictors

Key observations from the correlation analysis include:

- CVD Risk Score shows moderate correlation with BMI, LDL, cholesterol, and systolic blood pressure, indicating these variables may be important predictors.



EDA Key Insights

The exploratory analysis revealed several important patterns:

- The dataset contains diverse clinical, biochemical, and lifestyle indicators related to cardiovascular risk.
 - Missing values are relatively small (<6%) and manageable through preprocessing.
 - The dataset shows class imbalance, with high-risk patients being the most common.
 - Several health indicators show strong correlations, particularly BMI with weight and cholesterol with LDL.
 - CVD risk score is moderately associated with BMI, cholesterol, LDL, and systolic blood pressure, suggesting these may be important predictors.
- 

Modeling Approaches

Given the data, we had the option of using a classification model or a regression model (target labels: **CVD Risk Level vs. CVD Risk Score**)

- We chose to test both types of models and use the results to compare performance

3 types of models were chosen to approach this problem:

1. Gradient Boosting Classification Models (*LightGBM, XGBoost*)
2. Random Forest Classification Model
3. Gradient Boosting/Random Forest Regressors

Classification Modeling Implementation

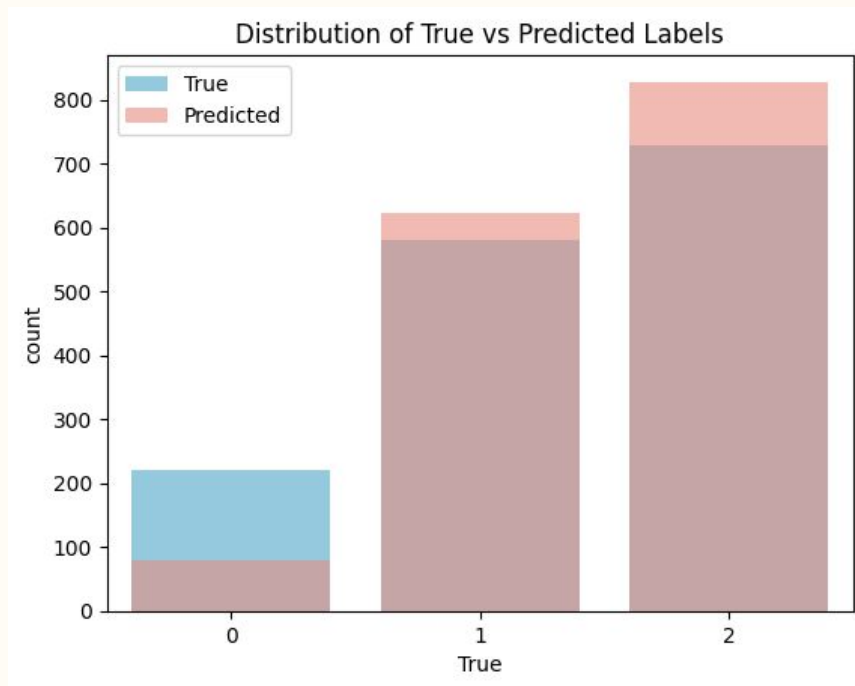
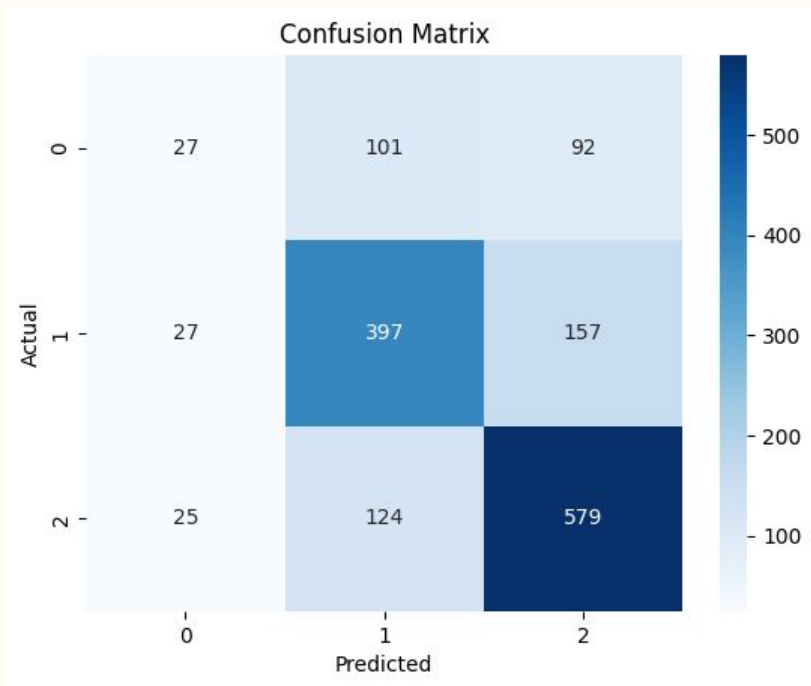
1. Data Preprocessing

- a. Encoding categorical values (physical activity, blood pressure, family history, smoking, diabetes)
- b. Removal of null/missing values from CVD Risk Level column
 - i. Tested removal of *Systolic BP* and *Diastolic BP*, but this lowered accuracy and R2 scores of all models tested

2. Model Building (XGBoost, LightGBM, Random Forest)

- a. **Stratified KFold split** with `n_splits = 10` (tested with 50 as well to verify results)

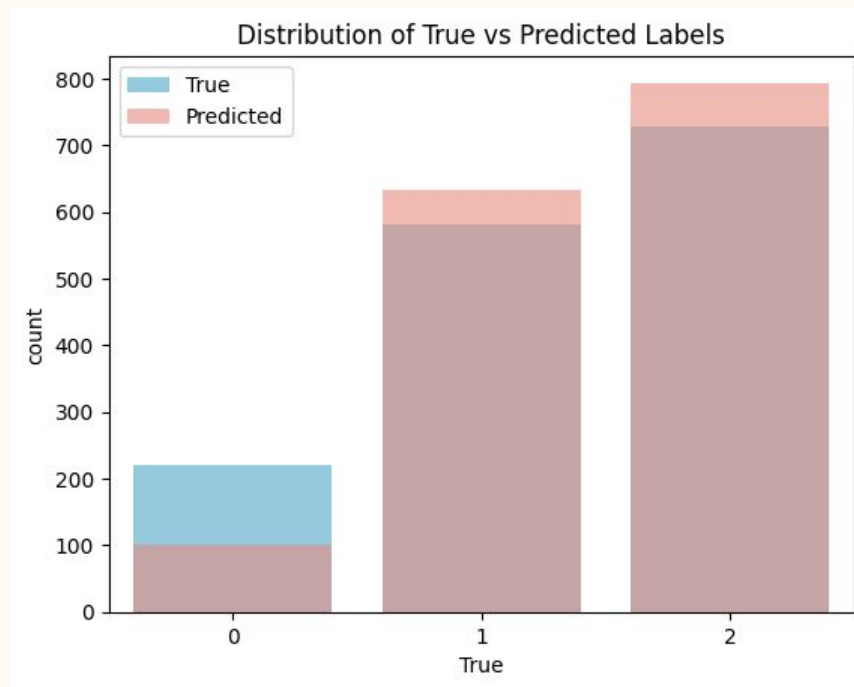
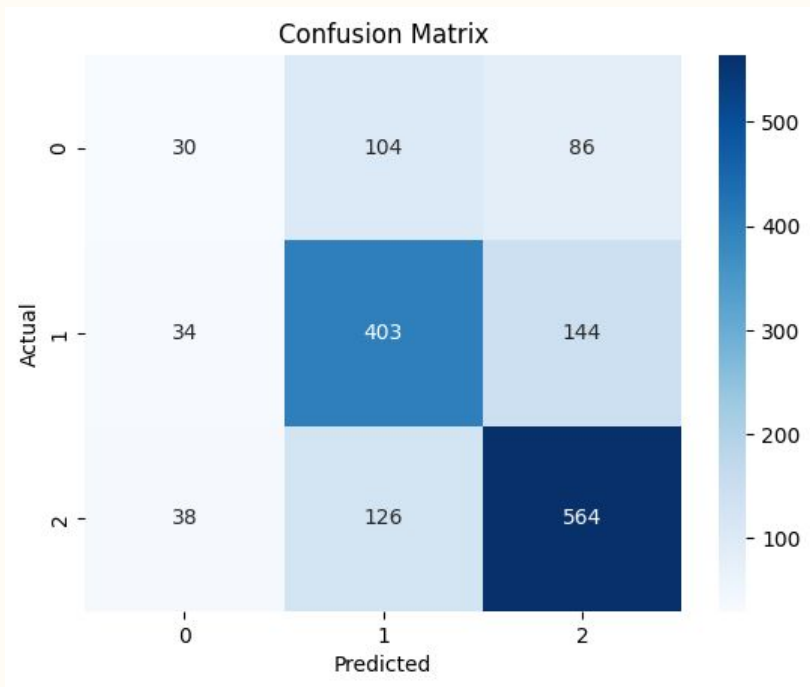
LightGBM Results



LightGBM Results

	precision	recall	f1-score	support
LOW	0.34	0.12	0.18	220
INTERMEDIARY	0.64	0.68	0.66	581
HIGH	0.70	0.80	0.74	728
accuracy			0.66	1529
macro avg	0.56	0.53	0.53	1529
weighted avg	0.62	0.66	0.63	1529

XGBoost Results

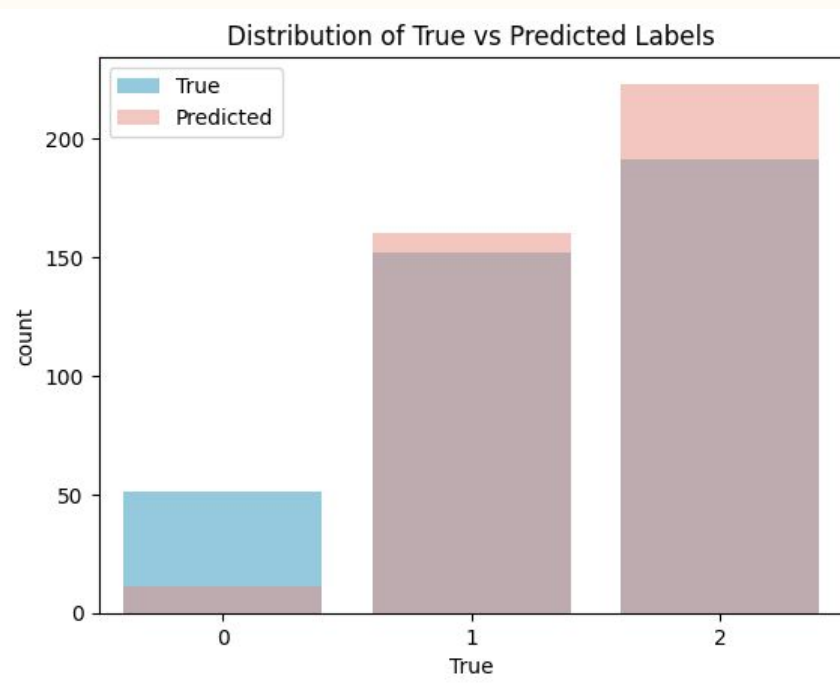
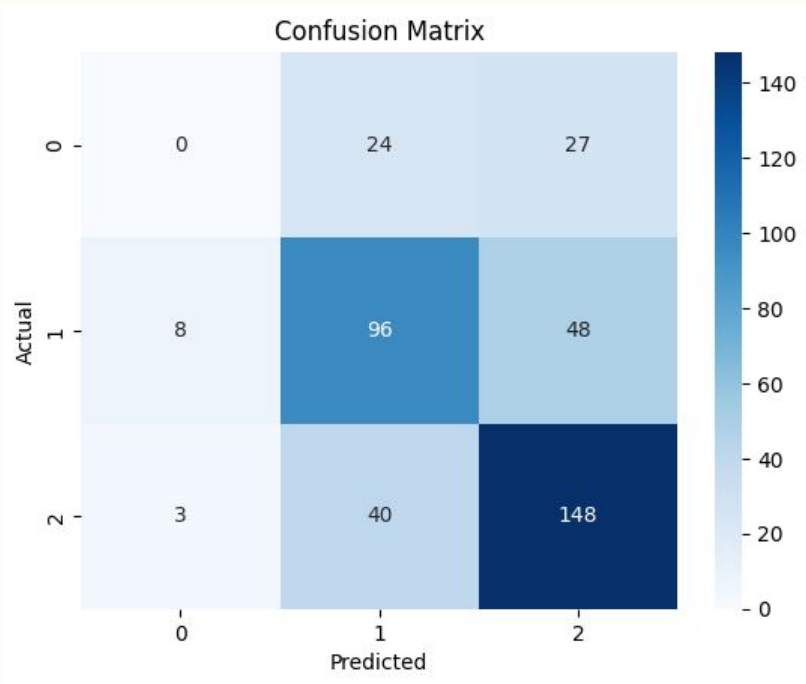


XGBoost Results

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HIGH	0.71	0.77	0.74	728
accuracy			0.65	1529
macro avg	0.55	0.53	0.53	1529
weighted avg	0.62	0.65	0.63	1529

Random Forest Results

Note: poor prediction of Low CVD Risk Level



Random Forest Results

	precision	recall	f1-score	support
LOW	0.00	0.00	0.00	51
INTERMEDIARY	0.60	0.63	0.62	152
HIGH	0.66	0.77	0.71	191
accuracy			0.62	394
macro avg	0.42	0.47	0.44	394
weighted avg	0.55	0.62	0.58	394

Classification Model Result Comparison

Baseline (zero-rule classifier): 0.48 (predicting High as the CVD Risk Level for every sample)

	LightGBM	XGBoost	Random Forest
<i>Accuracy</i>	0.66	0.65	0.62
<i>Weighted Average</i>	0.63	0.63	0.58

- LightGBM and XGBoost both performed better than RandomForest, but LightGBM performed slightly better than XGBoost for correctly classifying Low CVD Risk

Regression Modeling Implementation

1. Data Preprocessing

- a. Encoding categorical values (physical activity, blood pressure, family history, smoking, diabetes)
- b. Removal of null/missing values from CVD Risk Level column
 - i. Tested removal of Systolic BP and Diastolic BP, but lowered accuracy and R2 scores of all models tested

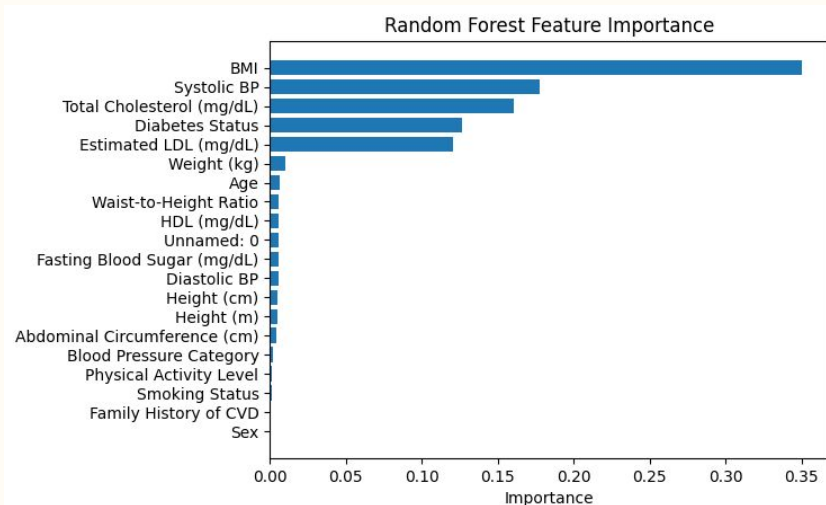
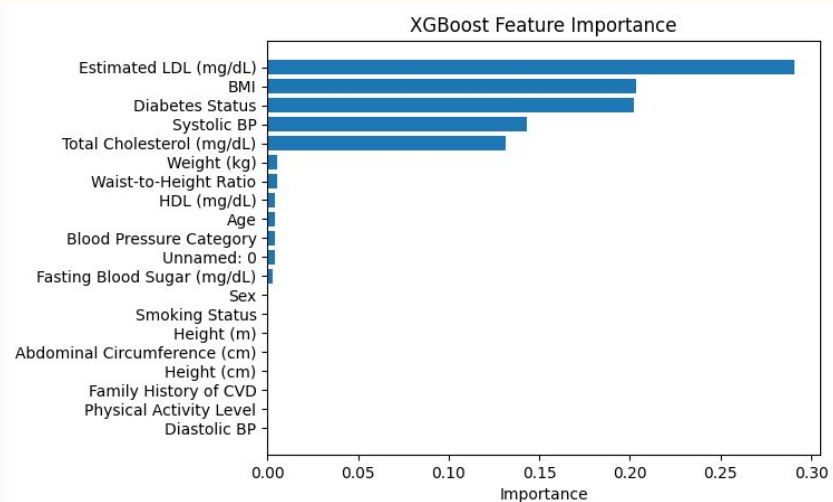
2. Data Scaling + Imputation

- a. usage of RobustScaler() and SimpleImputer() on numerical features

3. Model Building (*XGBoostRegressor*, *RandomForestRegressor*)

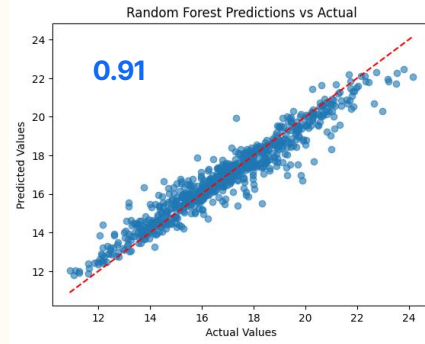
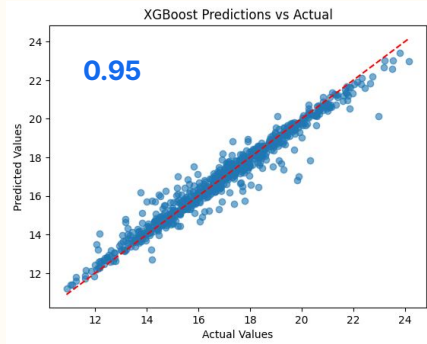
- a. Repetition to verify results (KFold split with 20 folds)

Model Feature Importance Comparison

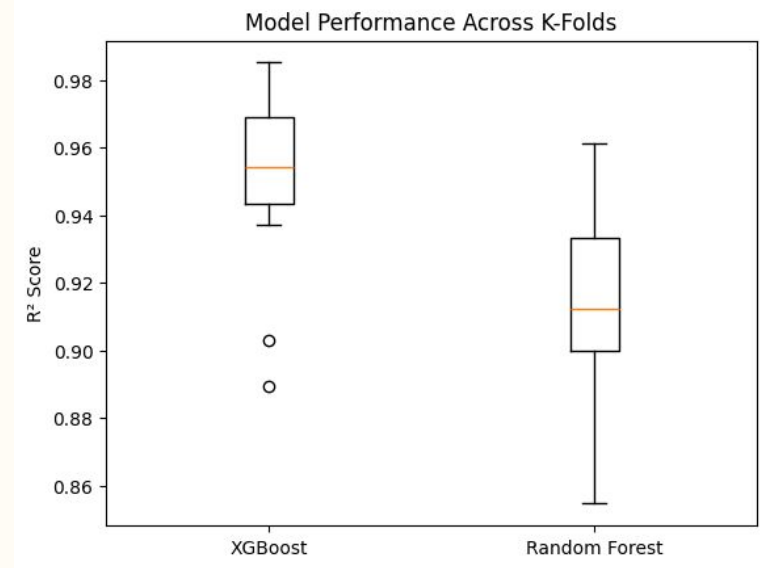


Each model highlights a *different feature* as the most important in determining CVD Risk Score, but they both **share the same top 5 most important features**.

XGBoostRegressor vs. RandomForestRegressor



Both regression models have a much higher accuracy rate than the classification models, but **XGBoostRegressor** outperforms **RandomForestRegressor** slightly



- A **linear relationship** between actual and predicted values is optimal → shows that predictions were almost the same as the actual values

Concluding Statements

1. The **regression models** used to predict based on CVD Risk Score performed much better than the **classification models** (based on CVD Risk Level)
 - a. despite using the same features and data preprocessing steps
 - b. had an average F1 accuracy rate of **0.95** from *XGBoostRegressor*
2. Feature importance validated the initial correlation matrix in that BMI, LDL, cholesterol, and diabetes status were the most impactful features

Future Work / Enhancements

1. Feature editing → removal of low correlation features (or features with low importance displayed by the bar chart) and multicollinearity (highly correlated variables)
 - a. explore other non-linear relationships between variables (ex: inclusion of Systolic BP and Diastolic BP in addition to Blood Pressure category)
2. Test using different scaling models and imputation procedures (MinMaxScaler, KNNImputer)
3. Investigate relationship when trained on a male patient dataset and tested on a female patient dataset (does this validate research?)



Questions?